Systemic Amplification in Modern Markets

Collateral Velocity, Funding Stress, and Microstructure-Controlled Price Discovery

with GameStop as a Structural Constraint Case Study

Research Working Paper

March 7, 2026

Abstract

Modern financial markets exhibit recurring instability patterns characterized by liquidity collapses, volatility clustering, discontinuous repricing, and cross-asset contagion. These outcomes are often insufficiently explained by changes in long-run fundamentals alone. This paper argues that modern instability is frequently generated by structural amplification mechanisms embedded within market architecture. Fragmented execution venues, synthetic exposure pathways, collateral reuse, procyclical clearing rules, and intermediary balance-sheet constraints interact to transform moderate disturbances into nonlinear price dynamics.

The paper introduces a mechanism-first framework in which market shocks propagate through a recursive loop: price movement increases volatility; higher volatility tightens margin and haircut conditions; tighter constraints force deleveraging; forced flows degrade liquidity; degraded liquidity raises market impact; and higher market impact further amplifies price movement. To operationalize these dynamics, the paper proposes a Systemic Amplification Index Proxy, or SAI-Proxy, a composite monitoring framework built from observable measures of funding stress, margin tightening, dealer capacity contraction, forced-flow intensity, and market impact.

GameStop, or GME, is used as a structural case study because it displays characteristics that make hidden system mechanics unusually visible: heavy derivative sensitivity, synthetic exposure elasticity, execution-fragmentation effects, and episodic constraint-driven repricing. The paper concludes with an empirical backtest framework, an operational data schema, Python implementation guidance, and a reform agenda focused on reducing margin procyclicality, improving exposure transparency, and strengthening systemic liquidity surveillance.

Keywords: systemic risk, market microstructure, collateral velocity, rehypothecation, funding stress, margin procyclicality, derivatives convexity, synthetic exposure, liquidity fragmentation, GameStop

1. Introduction

Financial instability increasingly manifests as liquidity failure rather than as classical insolvency recognition. Modern crises often emerge after extended periods of calm, compressed volatility, and apparently durable liquidity. The transition from stability to crisis is frequently abrupt. Markets that appear deep and continuous can rapidly become thin, impact-sensitive, and discontinuous.

Traditional equilibrium models struggle to explain these transitions because they often assume frictionless liquidity, relatively stable funding conditions, and price formation that primarily reflects information about fundamentals. In practice, modern financial markets are governed by a network of constraints. Funding availability, collateral eligibility, margin rules, execution routing, and intermediary inventory limits determine whether positions can be sustained and whether shocks can be absorbed. When these constraints tighten simultaneously, markets can move from a stable regime into an unstable regime even in the absence of proportionate changes in long-run valuation.

This paper advances a mechanism-driven framework for understanding these dynamics. The central claim is that modern instability is often generated by endogenous amplification loops embedded in the structure of collateralized and fragmented financial markets. The paper links the literatures on market microstructure, funding liquidity, margin procyclicality, and derivatives convexity into a unified framework and proposes a practical monitoring tool: the Systemic Amplification Index Proxy, or SAI-Proxy.

2. Claim Taxonomy and Evidentiary Standards

This paper classifies statements into three evidentiary tiers:

1. **Confirmed**: directly supported by primary datasets, regulatory filings, official institutional documentation, or established academic literature.

2. **Supported inference**: consistent with multiple observable proxies, historical episodes, and mechanism-consistent evidence, but not fully disclosed in primary form.

3. **Interpretive**: plausible and mechanism-consistent, but requiring additional transparency or future data to confirm.

Interpretive claims are accompanied by falsifiability criteria where feasible. This structure is intended to preserve analytic rigor while acknowledging that portions of modern market leverage remain only partially observable.

3. Literature Review

The framework developed here draws from four major research traditions.

First, the market microstructure literature shows that execution architecture affects both the location and speed of price discovery. Representative work includes Maureen O'Hara on modern microstructure, Albert Menkveld on high-frequency market making, and Budish, Cramton, and Shim on market design and latency competition.

Second, the literature on funding liquidity demonstrates that market liquidity and funding liquidity are jointly determined. The liquidity spiral framework developed by Brunnermeier and Pedersen is central to the present paper.

Third, research on post-crisis market structure and clearing procyclicality documents how risk control systems can amplify stress. Work by Darrell Duffie and by Murphy, Vasios, and Vause is especially relevant in understanding how constraints intensify during market stress.

Fourth, the options and volatility literature demonstrates that convexity can generate mechanical hedging flows that overwhelm directional beliefs in short windows. In derivatives-dense assets, gamma and vega exposures can transform modest spot moves into self-reinforcing hedging demand.

The present contribution is to integrate these literatures into a single systemic amplification model and then translate the model into an operational monitoring framework.

4. Historical Context: From Capital Buffers to Velocity Systems

Earlier financial systems were more balance-sheet centric. Banks absorbed risk, and crises often emerged through insolvency recognition or credit losses that accumulated gradually. Modern markets are more reliant on collateralized funding, derivatives intermediation, and market-based risk transfer. As a result, crises are increasingly transmitted through liquidity and funding constraints rather than through slow recognition of permanent loss.

This transition has several implications. First, time horizons compress. Participants who could survive a gradual repricing may not survive a rapid margin-driven deleveraging. Second, balance-sheet capacity becomes a macro variable. Dealer and intermediary willingness to warehouse risk now influences whether price discovery remains continuous. Third, leverage

becomes less visible. Exposure is often embedded in derivative structures, ETF baskets, swaps, and collateral reuse chains.

Historical episodes illustrate these dynamics. The 1987 crash showed how feedback trading can transform hedging logic into destabilizing flow. The LTCM crisis demonstrated that sophisticated relative-value portfolios can unravel when funding dries up and correlations converge. The 2008 crisis showed how leverage migrates into shadow channels when regulated bank leverage contracts. The March 2020 Treasury market episode demonstrated that even core collateral assets can become disorderly when they function simultaneously as safe assets and liquidation instruments.

5. Liquidity as an Emergent System Property

Liquidity is frequently described as if it were a fixed characteristic of an asset. This is analytically incomplete. Liquidity is better understood as an emergent regime condition that depends on the joint availability of stable funding, reusable collateral, willing intermediaries, and execution environments capable of distributing inventory without overwhelming visible order books.

When these conditions hold, markets appear deep, spreads are narrow, and price discovery seems smooth. When one or more fail, liquidity does not necessarily degrade linearly. It often fails discontinuously. Quoted depth evaporates, market impact increases nonlinearly, and displayed prices become increasingly sensitive to flow rather than to information.

This interpretation implies that market instability is frequently a *constraint event* rather than merely a narrative event. The relevant question is not simply what news arrived, but which constraints tightened, which positions became unfundable, and which flows became forced.

6. Core Mechanism: The Systemic Amplification Loop

The central mechanism proposed in this paper is a recursive feedback loop:

1. Price moves.
2. Volatility rises.
3. Margin and haircut requirements tighten.
4. Leverage capacity declines.
5. Forced liquidation increases.

6. Liquidity deteriorates and market impact rises.

7. Additional price movement occurs.

This loop is mechanical rather than conspiratorial. It does not require explicit coordination. It requires only that the system be leveraged, that margin models respond procyclically to volatility, and that liquidity provision depend on intermediary balance-sheet capacity.

The loop can be summarized compactly as

$$\Delta P \to \sigma \uparrow \to m \uparrow \to L \downarrow \to \Phi \uparrow \to \lambda \uparrow \to \Delta P \uparrow$$

where ΔP is price movement, σ is volatility, m is margin intensity, L is leverage capacity, Φ is forced flow, and λ is market impact.

7. Collateral Velocity and Rehypothecation

Collateral acts as the working capital of modern finance. It enables leverage, supports derivatives exposure, and sustains market-making. In calm conditions, collateral can be reused through rehypothecation chains and funding markets. The same unit of collateral may therefore support multiple layers of exposure. This creates *collateral velocity*.

Collateral velocity is efficient in stable conditions but fragile in stress conditions. If haircuts rise, if eligibility standards tighten, or if counterparties become more conservative, the same nominal collateral supports less effective funding. Because leverage is built on the post-haircut value of collateral rather than the nominal value alone, modest changes in haircut assumptions can create disproportionate deleveraging pressure.

This mechanism helps explain why crises can begin without widespread insolvency. Participants may remain fundamentally solvent but unable to sustain positions at the new financing terms. Under those conditions, forced selling becomes mandatory.

8. Funding Stress Transmission

Funding markets govern the feasible set of leverage. A participant may hold strong convictions about valuation, but cannot maintain exposure if financing becomes unavailable or too expensive.

Three channels are especially important:

1. **Repo markets**: secured financing for inventories and arbitrage books.

2. **FX basis and dollar funding channels**: external dollar scarcity that propagates into domestic risk assets.

3. **Carry structures**: embedded leverage supported by stable funding differentials and low volatility.

When funding conditions tighten, balance-sheet capacity contracts. Dealer inventories shrink, quote sizes fall, and the system becomes less able to absorb flow. In parallel, leveraged participants are forced to reduce positions. The resulting price movements often appear disconnected from fundamentals because they are driven by financing constraints rather than by directional beliefs.

9. Market Microstructure and Discovery Control Surfaces

Modern equity markets are fragmented across displayed exchanges, alternative trading systems, internalizers, and routing networks. This architecture creates a separation between public price references and private risk transfer.

One implication is that high volume can coexist with muted visible price discovery. If a substantial share of flow is internalized or matched off-exchange, the displayed order book may not fully reflect the economic pressure in the system. In stable regimes, this may improve execution efficiency. In stress regimes, it can defer visible discovery and allow inventory transfer to occur away from lit price formation.

This is one reason why persistent claims of "price management" arise in constrained assets. Even absent wrongdoing, the structure permits a divergence between economic pressure and displayed movement.

10. Synthetic Supply: ETFs, Swaps, and Options

Synthetic instruments increase the effective elasticity of supply. ETFs, swaps, and options can transfer economic exposure without requiring immediate cash-market settlement. This means that reported short interest may understate true systemic short pressure.

ETFs can provide effective share availability through basket and creation-redemption mechanics. Total return swaps allow economic short exposure to be warehoused in counterparty relationships. Options can create delta and gamma exposures that alter the timing and location of hedging demand.

The analytical consequence is important: visible spot metrics may not fully capture the degree of constraint embedded in a name. Observable short interest can become a lower bound rather than a complete measure.

11. GameStop as a Structural Constraint Case Study

GameStop is used as a case study not because it is unique in principle, but because it is unusually transparent as a stress sensor. It combines several characteristics that surface hidden mechanics:

1. limited effective float relative to trading activity,
2. significant derivative participation and concentrated option open interest,
3. persistent public debate over short exposure and synthetic positioning,
4. observable shifts in execution and liquidity regime behavior.

In a fundamentals-only framework, persistent and severe dislocations would be difficult to reconcile solely with valuation updates. In a constraint framework, however, such dislocations can persist if the system is primarily managing exposure pathways rather than repricing fundamental cash flows.

The case study is therefore illustrative. It does not require claims of coordination to be analytically useful. It only requires that the asset be sufficiently constrained for structural dynamics to become visible.

12. Gamma, Convexity, and Nonlinear Event Structure

Options are not only risk-transfer instruments. They are also flow generators. When market makers intermediate option demand, they often hedge dynamically in the underlying. The sign and magnitude of gamma determine whether these hedging flows dampen or amplify spot moves.

In a short-gamma regime, dealers buy into rising prices and sell into falling prices. This reinforces price movement and can create instability. In a long-gamma regime, hedging flows are stabilizing and mean-reverting.

These dynamics matter most when:

- option open interest is high relative to float,
- large strike concentrations sit near spot,

- implied volatility is unstable,
- and the underlying asset is already liquidity constrained.

Expiration windows compress these effects into narrow intervals. When exposures must be rolled or reset, deferred imbalances become harder to maintain. These windows therefore serve as time-compression points for systemic stress.

13. Formal Mathematical Model

We define the state vector

$$X_t = [P_t, \sigma_t, m_t, F_t, D_t, \Phi_t, \lambda_t]$$

where:

- P_t: price level,
- σ_t: volatility state,
- m_t: margin or haircut constraint intensity,
- F_t: funding stress,
- D_t: dealer or intermediary capacity,
- Φ_t: forced liquidation flow intensity,
- λ_t: market impact or liquidity state.

The update equations are

$$\Delta P_t = -\lambda_t \Phi_t + \epsilon_t$$

$$\sigma_{t+1} = \alpha_\sigma \sigma_t + \beta_\sigma |\Delta P_t| + \gamma_\sigma \lambda_t$$

$$m_{t+1} = \alpha_m m_t + \beta_m \sigma_{t+1}$$

$$F_{t+1} = \alpha_F F_t + \beta_F \sigma_{t+1} + \gamma_F m_{t+1}$$

$$D_{t+1} = D_t - \beta_D \sigma_{t+1} - \gamma_D F_{t+1}$$

$$\lambda_{t+1} = \alpha_\lambda \lambda_t + \beta_\lambda \left(\frac{1}{D_{t+1}} \right)$$

$$\Phi_{t+1} = \alpha_\Phi \Phi_t + \beta_\Phi m_{t+1} + \gamma_\Phi F_{t+1}$$

This system becomes unstable when the effective feedback gain exceeds the system's absorptive capacity. The exact gain condition can be estimated empirically through proxy calibration and backtesting.

14. The SAI-Proxy Framework

Because the latent state variables are not directly observable in full, this paper proposes an operational approximation: the Systemic Amplification Index Proxy.

The SAI-Proxy aggregates five measurable buckets:

1. Funding stress $F(t)$
2. Margin or haircut tightening $M(t)$
3. Dealer capacity contraction $D(t)$
4. Forced-flow pressure $\Phi(t)$
5. Market impact or liquidity degradation $\lambda(t)$

The composite score is defined as

$$SAI_t = w_1 F(t) + w_2 M(t) + w_3 D(t) + w_4 \Phi(t) + w_5 \lambda(t)$$

A baseline weighting scheme for operational use is

$$w_1 = 0.25, \quad w_2 = 0.20, \quad w_3 = 0.20, \quad w_4 = 0.20, \quad w_5 = 0.15$$

These weights are starting values and should be re-estimated using historical fit and out-of-sample predictive performance.

15. Operational Dataset Specification

The weekly SAI dataset should include the following columns:

Column	**Description**
date	Observation date, weekly frequency
ticker	Asset identifier
funding_stress	Composite funding stress signal
repo_spread	Repo or money-market stress proxy
fx_basis	Cross-currency basis or dollar scarcity indicator
margin_proxy	Composite margin pressure proxy
realized_vol	Realized volatility
implied_vol	Implied volatility proxy

dealer_capacity	Composite liquidity or intermediation proxy
spread_width	Bid-ask spread or equivalent spread measure
depth_proxy	Order book depth or depth surrogate
forced_flow	Composite forced-flow intensity measure
downside_volume_ratio	Share of downside volume in total volume
gap_intensity	Frequency or magnitude of discontinuous moves
market_impact	Range-to-volume or impact proxy
sai	Final normalized SAI-Proxy score
regime_label	Stable, Elevated, Unstable, or Nonlinear

Suggested Observable Inputs

- **Funding stress**: SOFR changes, repo proxies, FX basis, dollar scarcity indicators.
- **Margin stress**: VIX, realized volatility, volatility-of-volatility, dispersion metrics.
- **Dealer capacity**: bid-ask spreads, depth, off-exchange share, range relative to volume.
- **Forced flow**: gap frequency, clustered down moves, volume spikes on declines.
- **Market impact**: intraday range normalized by volume, slippage proxies, spread expansion.

16. Empirical Data Pipeline

The following Python sketch illustrates a base data pipeline for weekly SAI construction.

```
import pandas as pd
import numpy as np
import pandas_datareader.data as web
import datetime

start = datetime.datetime(2010,1,1)
end = datetime.datetime.today()

# Funding and volatility proxies
sofr = web.DataReader("SOFR", "fred", start, end)
vix = web.DataReader("VIXCLS", "fred", start, end)
```

```
df = pd.concat([sofr, vix], axis=1)
df.columns = ["sofr", "vix"]

df["funding_proxy_raw"] = df["sofr"].pct_change()
df["margin_proxy_raw"] = df["vix"].pct_change()

df = df.resample("W").mean()

def zscore(x):
    return (x - x.mean()) / x.std()

df["funding_proxy"] = zscore(df["funding_proxy_raw"])
df["margin_proxy"] = zscore(df["margin_proxy_raw"])

# Placeholder columns to be replaced with actual microstructure data
df["dealer_capacity_raw"] = np.nan
df["forced_flow_raw"] = np.nan
df["market_impact_raw"] = np.nan

# Final SAI score once all proxies are populated
df["sai"] = np.nan
```

In production use, this weekly base table should be joined with market microstructure and options data before the final normalization step.

17. Monte Carlo Simulation Framework

The theoretical model can be explored through Monte Carlo simulation. The objective is not exact price prediction, but stability analysis.

```
import numpy as np

T = 200
P = np.zeros(T)
sigma = np.zeros(T)
margin = np.zeros(T)
funding = np.zeros(T)
dealer = np.ones(T)
```

```
impact = np.zeros(T)
flow = np.zeros(T)

for t in range(1, T):
    sigma[t] = 0.8 * sigma[t-1] + 0.4 * abs(P[t-1])
    margin[t] = 0.7 * margin[t-1] + 0.5 * sigma[t]
    funding[t] = 0.6 * funding[t-1] + 0.4 * margin[t]
    dealer[t] = max(0.1, dealer[t-1] - 0.3 * sigma[t])
    impact[t] = 0.5 * impact[t-1] + (1 / dealer[t])
    flow[t] = 0.5 * margin[t] + 0.5 * funding[t]
    P[t] = P[t-1] - impact[t] * flow[t] + np.random.normal(0, 0.02)
```

Simulations typically generate three regimes: stable, elevated, and unstable. The unstable regime is characterized by convex price paths, persistent volatility, and sharp deterioration in effective dealer capacity.

18. Empirical Regression Framework

The paper proposes three core empirical tests.

18.1. Forward Volatility Regression

$$RV_{t+k} = \alpha + \beta_1 SAI_t + \beta_2 VIX_t + \beta_3 RV_t + \epsilon_t$$

where RV_{t+k} is realized volatility over a forward horizon k, SAI_t is the systemic amplification index, VIX_t is implied volatility, and RV_t is current realized volatility.

The central hypothesis is

$$\beta_1 > 0$$

meaning that elevated SAI values predict higher forward realized volatility beyond what is already explained by baseline volatility measures.

18.2. Drawdown Probability Model

$$Pr(Drawdown_{t+k} = 1) = \frac{1}{1 + e^{-(\alpha + \beta SAI_t + \gamma Z_t)}}$$

where the dependent variable equals 1 if the asset or index experiences a drawdown beyond a chosen threshold within k periods, and Z_t is a vector of controls.

18.3. Liquidity Stress Regression

$$Impact_{t+k} = \alpha + \beta_1 SAI_t + \beta_2 Spread_t + \beta_3 Vol_t + \epsilon_t$$

This model tests whether SAI predicts future market impact, controlling for current spread and volatility levels.

19. Empirical Crisis Backtests

To evaluate the descriptive and predictive usefulness of the SAI framework, the paper proposes event-window backtests across major crisis episodes.

Table 2: Historical Crisis Backtest Windows

Event	**Approximate Date**	**Expected Mechanism Signature**
Flash Crash	May 2010	liquidity collapse, impact spike
China Volatility Shock	Aug 2015	cross-asset stress, volatility expansion
Volmageddon	Feb 2018	convexity unwind, forced flow
COVID Liquidity Crisis	Mar 2020	funding stress, broad deleveraging
Treasury Market Dysfunction	Mar 2020	collateral stress, dealer retreat
Regional Bank Stress	Mar 2023	funding and confidence transmission

The backtest procedure is:

1. Construct weekly SAI-Proxy scores over a multi-year sample.
2. Normalize each component using rolling z-scores or percentile ranks.
3. Compare SAI levels in event windows with non-event periods.
4. Test whether elevated SAI values lead realized volatility spikes and drawdown frequency.

20. Illustrative Regression Results Template

The table below is a manuscript-ready template. Values shown are placeholders and must be replaced by actual estimates after running the empirical model.

Table 3: Illustrative Predictive Regression Results

Model	**Dependent Variable**	β_{SAI}	**Interpretation**
Model 1	Forward realized volatility	0.42	positive association
Model 2	Drawdown probability	0.31	positive association
Model 3	Future market impact	0.37	positive association

These values are provided as layout placeholders only. They are not empirical findings unless estimated from actual data.

21. Figure Placeholders

Figure 1 Placeholder
Conceptual systemic amplification loop

Figure 1: Conceptual systemic amplification loop.

Figure 2 Placeholder
Stylized SAI regime states

Figure 2: Illustrative SAI path across stable, elevated, unstable, and nonlinear regimes.

Figure 3 Placeholder
Funding stress versus forward volatility

Figure 3: Stylized positive relationship between funding stress and future volatility.

Figure 4 Placeholder
Margin procyclicality illustration

Figure 4: Stylized nonlinear increase in margin requirements as volatility rises.

Figure 5 Placeholder
Dealer capacity compression

Figure 5: Illustrative contraction in dealer risk-bearing capacity across stress regimes.

Figure 6 Placeholder
Market impact convexity

Figure 6: Stylized convex market impact as liquidity deteriorates.

Figure 7 Placeholder
Synthetic supply pathways

Figure 7: Illustrative pathways through which synthetic instruments increase effective supply elasticity.

Figure 8 Placeholder
Constraint asset event structure

Figure 8: Stylized constraint-asset event structure with stress leading discontinuous repricing.

Figure 9 Placeholder
Options convexity example

Figure 9: Stylized dealer hedge demand in a short-gamma regime.

Figure 10 Placeholder
Crisis backtest event map

Figure 10: Illustrative SAI spikes around crisis windows.

Figure 11 Placeholder
Volatility prediction illustration

Figure 11: Stylized monotonic increase in forward volatility across SAI deciles.

Figure 12 Placeholder
Drawdown probability illustration

Figure 12: Stylized logistic relationship between SAI and drawdown probability.

Figure 13 Placeholder
SAI component heat map

Figure 13: Qualitative component heat map. Replace with empirical heat map after data construction.

Figure 14 Placeholder
SAI dashboard mockup

Figure 14: Illustrative SAI dashboard output.

Figure 15 Placeholder
End-to-end system architecture

Figure 15: System architecture linking funding, collateral, intermediation, execution, and price discovery.

22. Operational Python Model

The following end-to-end Python scaffold combines normalization, SAI construction, and regime labeling.

```
import pandas as pd
import numpy as np

def zscore(series):
    return (series - series.mean()) / series.std()

def compute_sai(df):
    df = df.copy()

    df["funding_norm"] = zscore(df["funding_stress"])
    df["margin_norm"] = zscore(df["margin_proxy"])
    df["dealer_norm"] = zscore(df["dealer_capacity"])
    df["forced_norm"] = zscore(df["forced_flow"])
    df["impact_norm"] = zscore(df["market_impact"])

    df["sai"] = (
```

```
        0.25 * df["funding_norm"] +
        0.20 * df["margin_norm"] +
        0.20 * df["dealer_norm"] +
        0.20 * df["forced_norm"] +
        0.15 * df["impact_norm"]
    )

    def regime_label(x):
        if x < 1.0:
            return "Stable"
        elif x < 2.0:
            return "Elevated"
        elif x < 3.0:
            return "Unstable"
        return "Nonlinear"

    df["regime_label"] = df["sai"].apply(regime_label)
    return df
```

23. Falsifiability Criteria

The systemic amplification framework would be materially weakened if the following empirical results fail to hold:

1. SAI does not improve forward volatility prediction relative to naive or baseline volatility models.
2. SAI does not correlate with future liquidity deterioration or market impact.
3. Funding stress and margin proxies do not co-move during known stress windows.
4. Crisis episodes do not exhibit elevated SAI levels relative to non-crisis baselines.
5. Microstructure and synthetic exposure variables add no explanatory power in constrained assets.

If these failures occur, the model should be treated as descriptive rather than predictive, and the weighting or proxy design should be revised.

24. Policy Implications

Three policy implications follow directly from the framework.

First, margin frameworks should be reviewed for procyclicality. A system that tightens aggressively into volatility can transform risk control into risk amplification.

Second, transparency around synthetic exposures should be improved. Swaps, ETF transformations, and options-driven effective short exposure can materially change the system's constraint structure without appearing clearly in traditional short-interest statistics.

Third, regulators and market operators should maintain systemic liquidity dashboards that integrate funding, margin, execution, and impact measures rather than relying on price or volatility alone.

25. Conclusion

Modern financial markets are increasingly governed by mechanical feedback systems rather than by simple informational equilibria. Collateral velocity, fragmented execution, synthetic exposure, derivatives convexity, and procyclical margin systems can jointly create a structural amplification engine.

The SAI-Proxy proposed in this paper offers a practical approach to monitoring those dynamics. It is designed not as a perfect measure of hidden leverage, but as an operational framework for detecting when the system is moving from stability toward amplification.

GameStop serves as a useful case study because it makes these hidden dynamics easier to observe, but the broader framework applies to modern markets more generally. The central implication is that many apparently anomalous price events are best understood not as isolated curiosities, but as manifestations of system architecture operating under stress.

A. Appendix A: Example Weekly Dataset Row

Table 4: Illustrative Weekly Dataset Example

date	ticker	funding	margin	dealer	forced	impact	sai
2026-03-01	GME	1.21	1.76	-0.88	1.34	1.59	1.15

B. Appendix B: Backtest Code Sketch

```
# Example backtest logic
```

```
df["forward_vol"] = df["realized_vol"].shift(-4)

# OLS style:
# forward_vol ~ sai + implied_vol + realized_vol

# Logistic drawdown example:
# drawdown_flag = 1 if next_4w_return < threshold else 0
```

C. Appendix C: Notes on Replacing Placeholder Figures

All figures included in this manuscript are placeholder boxes designed to make the paper compile cleanly on limited editors. Replace them later with exported PDF, PNG, or JPG figures as empirical work is completed.

References

1. Brunnermeier, Markus K., and Lasse Heje Pedersen. 2009. "Market Liquidity and Funding Liquidity." *Review of Financial Studies* 22, no. 6: 2201–2238.

2. Budish, Eric, Peter Cramton, and John Shim. 2015. "The High-Frequency Trading Arms Race: Frequent Batch Auctions as a Market Design Response." *Quarterly Journal of Economics* 130, no. 4: 1547–1621.

3. Duffie, Darrell. 2018. "Post-Crisis Bank Regulations and Financial Market Liquidity." *BIS Working Papers*, No. 720.

4. Menkveld, Albert J. 2013. "High Frequency Trading and the New-Market Makers." *Journal of Financial Markets* 16, no. 4: 712–740.

5. Murphy, David, Michalis Vasios, and Nicholas Vause. 2016. "An Investigation into the Procyclicality of Risk-Based Initial Margin Models." *Bank of England and Financial Stability Papers*.

6. O'Hara, Maureen. 2015. *High Frequency Market Microstructure*. Wiley.

7. Bank for International Settlements. *BIS Quarterly Review*, various issues on funding liquidity, collateral, market functioning, and dealer balance-sheet constraints.

8. International Monetary Fund. *Global Financial Stability Report*, various issues on leverage, non-bank financial intermediation, funding stress, and systemic liquidity.